INJECTIVITY OF THE CANONICAL MAP TO THE PUSHFORWARD OF A FLAT PULLBACK WHEN THE RELATIVE ASSOCIATED POINTS LIE IN THE IMAGE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Base change behaviour of the relative assassin, and vanishing of a module with no weakly associated points. Divisors, labels divisors-lemma-base-change-relative-assassin (proof line 6615), divisors-remark-base-change-relative-assassin (proof line 6616), divisors-lemma-weakly-ass-zero (proof line 6623) and divisors-lemma-ses-weakly-ass (proof line 6626). The first two show that the hypothesis that the relative associated points lie in the image is stable under base change; the third reduces injectivity to the vanishing of the weakly associated points of the kernel, and the fourth places those inside the weakly associated points of the module itself.
- Weakly associated points of a flat finite type module lie in the relative assassin. Same chapter flat.tex, label lemma-bourbaki-finite-type-general-base, invoked at proof line 6628 and claimed separately as S14-algebra-bourbaki-general-base. Shows that for a flat finite type module every weakly associated point is a relative associated point, so the hypothesis controls all the weakly associated points of the kernel.
- Faithful flatness of a flat local ring map and its consequence of universal injectivity. Algebra, label algebra-lemma-faithfully-flat-universally-injective, invoked at proof line 6641. Makes the map obtained by tensoring the stalk of the module with the local ring upstairs injective, which is what forces the stalk of the kernel to vanish at every relevant point.

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Source reference. Original lemma 05CK: sources/stacks-chow/tags.txt, source lines 6923–6923. Navigation pointer for \ref{algebra-lemma-faithfully-flat-universally-injective} used inside the counted span (line 6641); the argument that consumes it is retained in the span.

Source reference. Original lemma 05DC: sources/stacks-chow/tags.txt, source lines 6950–6950. Navigation pointer for \ref{divisors-lemma-base-change-relative-assassin} used inside the counted span (line 6615); the argument that consumes it is retained in the span.

Source reference. Original lemma 05AN: sources/stacks-chow/tags.txt, source lines 6856–6856. Navigation pointer for \ref{divisors-lemma-ses-weakly-ass} used inside the counted span (line 6626); the argument that consumes it is retained in the span.

Source reference. Original lemma 05AP: sources/stacks-chow/tags.txt, source lines 6857–6857. Navigation pointer for \ref{divisors-lemma-weakly-ass-zero} used inside the counted span (line 6623); the argument that consumes it is retained in the span.

Source reference. Original remark 05KL: sources/stacks-chow/tags.txt, source lines 7205–7205. Navigation pointer for \ref{divisors-remark-base-change-relative-assassin} used inside the counted span (line 6616); the argument that consumes it is retained in the span.

Source reference. Original lemma 05IJ: sources/stacks-flat/pinned-original-source.tex, source lines 4079–4079. Navigation pointer for \ref{lemma-bourbaki-finite-type-general-base} used inside the counted span (line 6628); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $g : X' \to X$ be a flat morphism of schemes over S with X locally of finite type over S. Let $\mathcal{F}$ be a finite type quasi-coherent $\mathcal{O}_X$-module which is flat over S. If $Ass_{X/S}(\mathcal{F}) \subset g(X')$ then the canonical map*

$$\mathcal{F} \longrightarrow g_*g^*\mathcal{F}$$

is injective, and remains injective after any base change.

Proof. The final assertion means that $\mathcal{F}_T \to (g_T)_*g_T^*\mathcal{F}_T$ is injective for any morphism $T \to S$. The assumption $\mathrm{Ass}_{X/S}(\mathcal{F}) \subset g(X')$ is preserved by base change, see Divisors, Lemma 05DC and Remark 05KL. The same holds for the assumption of flatness and finite type. Hence it suffices to prove the injectivity of the displayed arrow. Let $\mathcal{K} = \mathrm{Ker}(\mathcal{F} \to g_*g^*\mathcal{F})$. Our goal is to prove that $\mathcal{K} = 0$. In order to do this it suffices to prove that $\mathrm{WeakAss}_X(\mathcal{K}) = \emptyset$, see Divisors, Lemma 05AP. We have $\mathrm{WeakAss}_X(\mathcal{K}) \subset \mathrm{WeakAss}_X(\mathcal{F})$, see Divisors, Lemma 05AN. As $\mathcal{F}$ is flat we see from Lemma 05IJ that $\mathrm{WeakAss}_X(\mathcal{F}) \subset \mathrm{Ass}_{X/S}(\mathcal{F})$. By assumption any point x of $\mathrm{Ass}_{X/S}(\mathcal{F})$ is the image of some $x' \in X'$. Since g is flat the local ring map $\mathcal{O}_{X,x} \to \mathcal{O}_{X',x'}$ is faithfully flat, hence the map

$$\mathcal{F}_x \longrightarrow g^*\mathcal{F}_{x'} = \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{O}_{X',x'}$$

is injective (see Algebra, Lemma 05CK). This implies that $\mathcal{K}_x = 0$ as desired. □